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ENG 2003

Assignment 4

April 6, 2026

Group T4

Transport & Climate

Jonathan Hutchinson

Breakdown

1. Goals and Purpose
2. Section 1 - Ideal Scenario vs. Reality
3. Section 2 - The Battery & Direct Alternatives
4. Section 3 - Public Transit & Urban Planning
5. Discussion
6. Summary
7. References
8. AI Usage
9. Q & A

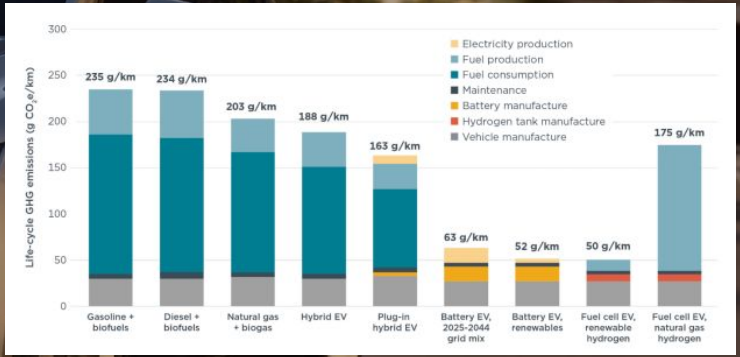
Goals and Purpose

- BBC’s 50 grand challenges
- BEV’s = Anti- Green Technology
- Public Transportation a Priority
- PHEV’s & Efficient City Design

1. The ideal scenario versus the reality

On Paper: BEVs Can Be Much Lower Carbon

- Zero tailpipe emissions during use
- With a fully renewable electricity grid:
  - lifecycle emissions can be much lower, up to 78%
- Lower use phase emissions can offset higher manufacturing emissions
  - ~17,000km cited
- Higher energy efficiency and lower operating costs

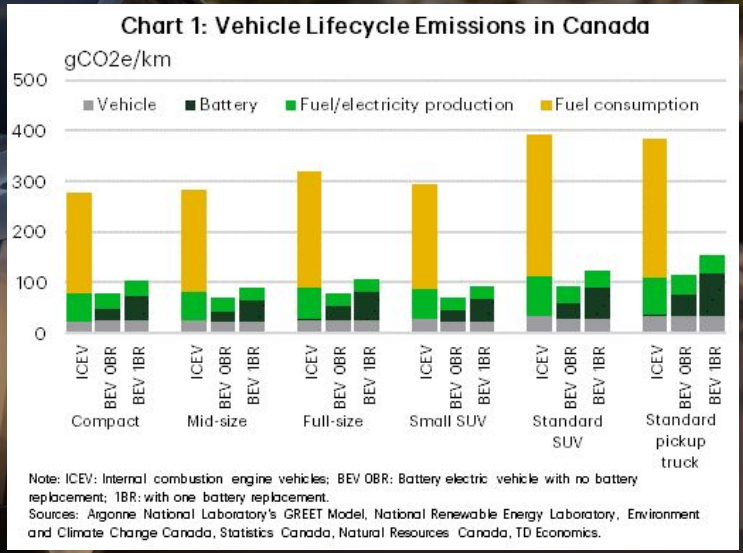


Neal Kaushal

Reality Check: Emissions Can Shift Upstream

- Manufacturing is energy intensive, creating an upfront “carbon debt”
- Cited that BEV manufacturing emissions ~40% higher than comparable vehicles
- Speed of paying off “carbon debt” depends largely on electricity mix and driving distance

Takeaway: *“clean” depends on the full lifecycle, not just tailpipe*



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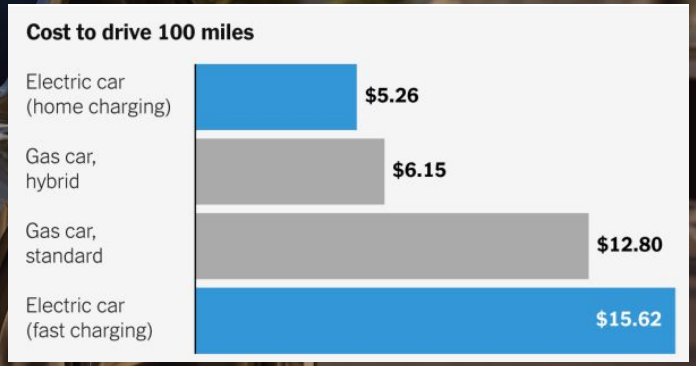
Cost and Convenience

- EV operating costs can be low with home charging
  - savings can shrink with public fast charging
- Fast charging can be much more expensive, rivaling or even potentially exceeding gas
- Public charging availability and pricing can easily become a bottleneck for fair adoption



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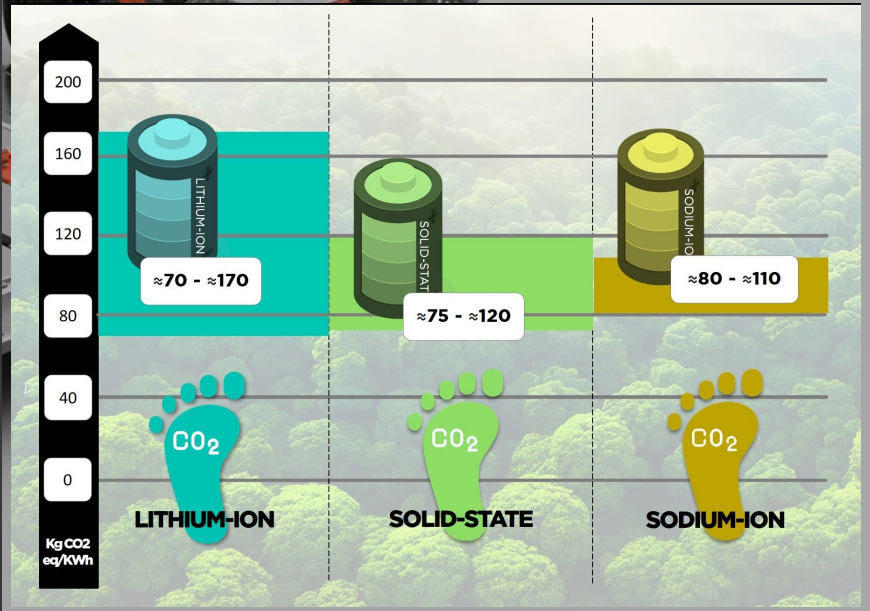
2. The Battery and Direct Alternatives

Reality of Batteries

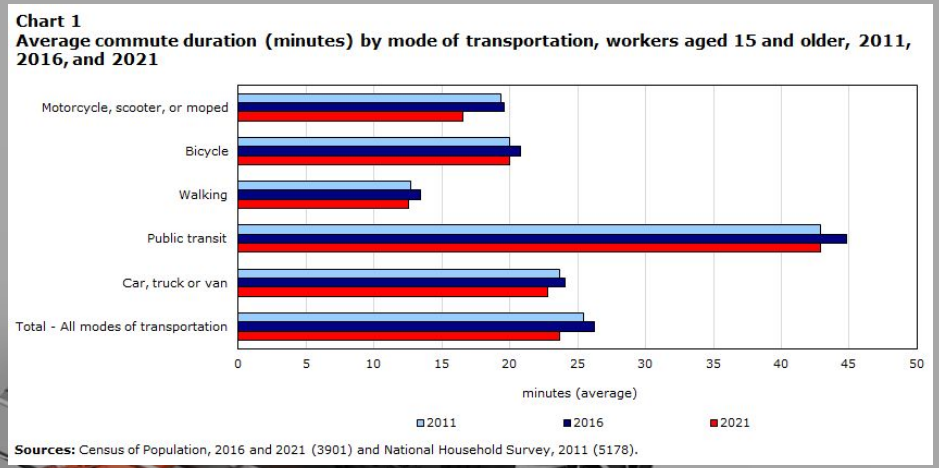
- Battery makes up 30-50% of emissions
- Require 15 tonnes of water on average
- Needs to be driven 20-50k km to break even
- Other major environmental damages

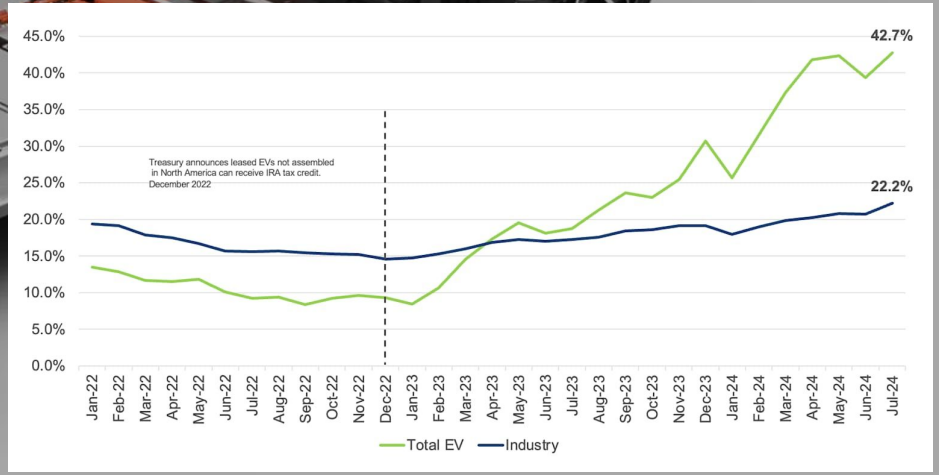


Biswaprakash Nayak

Alternatives

- Downsides of hydrogen
- Hybrids bridge the gap
- Average daily trips is around 25 mins by car
- 35-47% cars are leased, making it worse

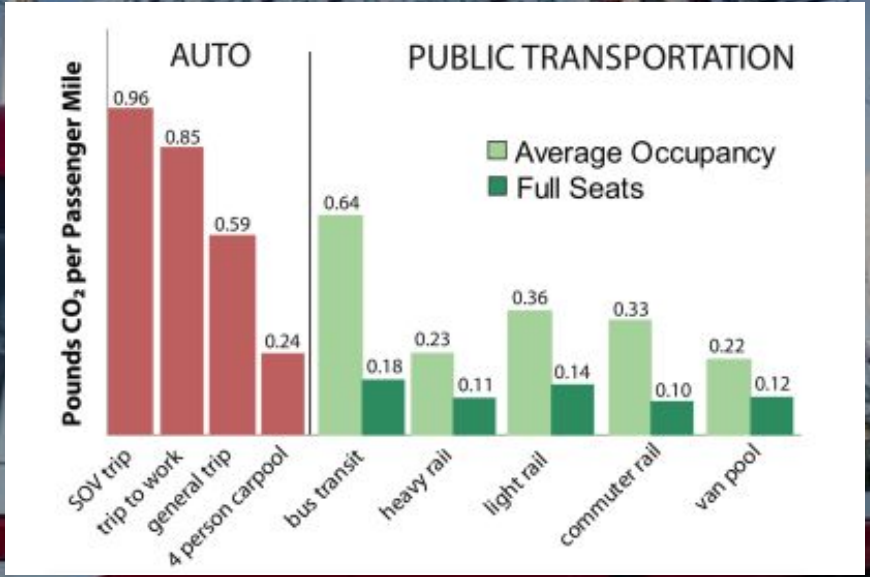




3. Public Transit and Urban Planning

Challenges in Public Transit Systems

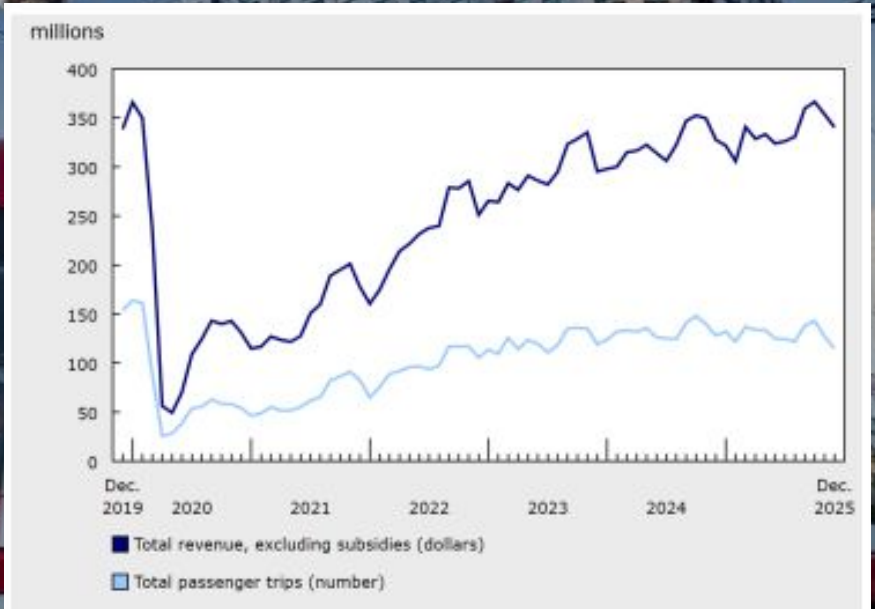
- Congestion and overcrowding in major cities
- Limited coverage in suburban areas
- Delays and inefficiencies in scheduling
- High dependence on cars due to gaps in transits



Sami Abou Asali

Urban Planning Solutions

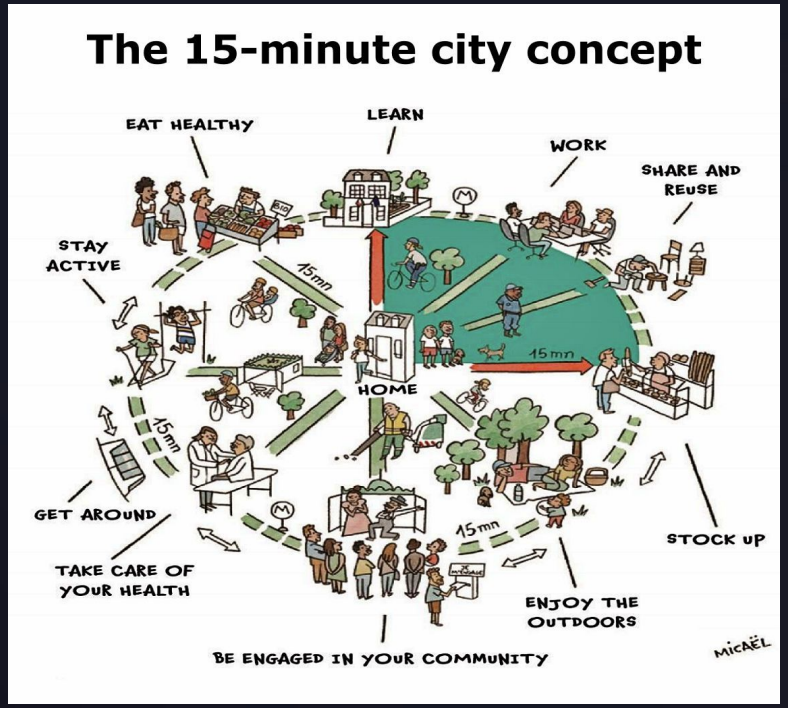
- Transit oriented development (TOD)
- Smart technologies (AI, realtime data)
- Expansion of transit networks
- Sustainable city designs



Sami Abou Asali

Discussion

- BEVs = Spreadsheet Lifetime Emission Savings
- City Service & Infrastructure Design
- Electric Transit
- Carbon Neutrality



https://johndecember.com/places/people/moreno2024-15-minute-city.jpg

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Summary

- BEV’s = High Carbon Footprint
- Hybrid and PHEV’s = The Future
- Public Transit & 15 Minute Cities
- Supportive Governmental Policies

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Thank You

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Ai Use (from the report)

In this minimal Ai has been used. I used perplexity mainly to hand it my points and then it found the sources I needed with a brief rundown on the source. I took these sources and read each of them and in the end chose what to keep and what to exclude. I overlooked each article and then created a generalized summary from them using it. The main use of perplexity was to give it my possible points of arguments which I got from generally searching around and It gave me related sources. I also gave it the outline so it knew what to look for and gave me more up to date articles and preferred more research papers than general articles but in the end I chose a few general articles too as they seemed to get the points across and had credible sources. I also asked it to create a draft for points I gave it so I could word it better but then I rewrote it entirely keeping parts I liked and removing unnecessary parts. The final version is my own writing.